Chapter 6, Problem 39P

Problem

The voltage across a 50-mH inductor is given by

$$v(t) = [5e^{-2t} + 2t + 4] \text{ V} \qquad \text{for } t > 0.$$

Determine the current $i(t)$ through the inductor. Assume that $i(0) = 0$ A.

Step-by-step solution

Step 1 of 2

Voltage across inductor is

$$v(\text{t}) = \left[5\text{e}^{-2\text{t}} + 2\text{t} + 4\right] \text{V}$$

Current through inductor is

$$i = \frac{1}{L}\int_0^t v(t)\, dt + i(0)$$

Substitute 50 mH for L and $\left[5\text{e}^{-2\text{t}} + 2\text{t} + 4\right] \text{V}$ for $v(t)$ and 0 A for $i(0)$.

$$= \frac{1}{50 \times 10^{-3}} \int_0^t \left(5e^{-2t} + 2t + 4\right) dt + 0$$

Step 2 of 2

Evaluate the integral to get the expression for current .

$$= 20\left(\frac{5}{-2}e^{-2t} + t^2 + 4t\right)\Big|_0^t$$

$$= -50e^{-2t} + 20t^2 + 80t - (-50)$$

$$i = -50e^{-2t} + 20t^2 + 80t + 50 \text{ A}$$

Current through 50 mH inductor is $\boxed{-50e^{-2t} + 20t^2 + 80t + 50 \text{ A}}$.